**Technical Report**

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**Introduction:**

Remote learning has earned its place as an operable method of instruction in the modern age, but many authorities have raised questions or concerns about its use in the face of COVID-19. As a group, we investigated the relationship between prominently-online schooling and student performance in higher academia to see if one exists. This investigation gave us a better understanding of how distance learning can foster (or hinder) students in the face of crises like COVID-19. The results from this project can be used to identify if distance learning is beneficial for students and what strategies perform best.

**Data:**

We used a dataset derived from the IPEDS (Integrated Postsecondary Education Data System) with features related to our problem. Our dataset accounts for public four-year institutions in the United States (approximately 571 institutions) across multiple years. The dataset includes data to identify institutions, institution demographics, and data about the presence of fully remote learning options.

The dataset includes the following features:

- unitid - unique institutional ID number
- institution_name - name of the college or university
- year - year the data was recorded
- avg_net_price - average net price of attendance for students
- grad_rate_150pct_4yr - 4-year institution graduation rate measured at 150% of normal completion time
- hispanic_count - number of Hispanic students enrolled
- black_count - number of Black students enrolled
- white_count - number of White students enrolled
- hispanic_share - proportion of enrolled students who are Hispanic
- black_share - proportion of enrolled students who are Black
- white_share - proportion of enrolled students who are White

- distance_only_count - number of students enrolled exclusively in distance education
- distance_only_share_imputed - imputed proportion of students enrolled exclusively in distance education
- distance_data_available - indicator for whether distance education data is available
- control_of_institution - control type of the institution
- level_of_institution - institution level
- sector_of_institution - institutional sector classification

Variable Notes

- grad_rate_150pct_4yr: Because this is measured at 150% of normal completion time. This means completion within about 6 years, not 4 years.
- avg_net_price: Represents average net cost after financial aid
- In general, variables related to distance education only count students who are fully remote learners. Students who take a mix of in-person and remote classes are not represented in these columns.

**Preprocessing and Exploratory Data Analysis:**

Prior to preprocessing, some basic exploratory data analysis was done to get a better idea of the data we were working with. We looked at the first 5 rows to ensure everything was read successfully, like the statistics behind the numerical data, the size of the dataset, variable types, etc.

After loading the data any duplicate rows were dropped along with columns that were consistent across the dataset. This included “control_of_institution,” “level_of_institution,” and “sector_of_institution.” These were consistent across the dataset because all colleges included were 4 year public colleges. After this, we checked for null values. Two rows had missing values, “distance_only_share_imputed” and “distance_data_available.” Because “distance_data_available” was missing two-thirds of its values, and because it would not be very useful for the model, it was also dropped. To fill in the missing values for “distance_only_share_imputed” a new variable was calculated for total enrollment. This was done by using the race count variables and the race share variables to find the total number of enrolled students. A few rows were confirmed by searching up their enrollment in the respective

years. Using this new variable, “distance_only_share_imputed” could be calculated using “distance_only_count.” Finally the 4 variables with counts were dropped as the percentage of students of a certain race was a better metric for the model than the number of students of that race. Finally all numerical variables were standardized.

After the dataset was cleaned some more exploratory data analysis was performed. We looked at the distribution of the target variable “grad_rate_150pct_4yr” (6 year graduation rates), and analyzed the relationship between features with a pairplot and a correlation heatmap (as seen in Figure 1). Additionally, since one of our questions was how much distance learning affected graduation rates we looked at both the graduation rates and percentage of distance students in the 3 years our dataset included. There was a marginal increase in graduation rates in 2020 compared to 2017 and 2023 (Figure 2) and a substantial increase in distance learning students in 2020 compared to the other two years (Figure 3). This led us to believe that an increase in distance learning students leads to higher graduation rates. Finally we wanted to look at the number of HBCUs present in the dataset. To do this we found the number of rows where the percentage of black students was higher than 80%. 200 colleges in our dataset fit this criteria.

**Modeling:**

The dataset was first divided into two groups: 80% for training and 20% for testing. Six models were created:

- Model 1: Multiple Regression Model without Race Share Variables
- Model 2: Multiple Regression Model with Race Share Variables
- Model 3: Basic Decision Tree Regressor
- Model 4: Decision Tree Regressor with Hyperparameter Tuning
- Model 5: Ridge Regression
- Model 6: Random Forest Regressor

Each model was trained on the same training subset and evaluated on the same testing subset to ensure a fair comparison. To measure performance, mean squared error, mean absolute error, root mean square error, and $R^2$ were calculated with $R^2$ being our main metric for evaluation.

The first two models were basic linear/multiple regression models, one of which did include race and one that did not. These two were compared to see if including race increased model accuracy. The second model performed better, so race was included in future models. The

third model was a basic decision tree regressor without any specific parameters. This was used as a baseline to compare to the hypertuned decision tree regressor that was created for model 4. The parameters tested in the second model were:

- Depth: the depth the decision tree will go to
- Criterion: whether the model will use squared error, friedman mse, or absolute error
- Splitters: whether the model will choose the best split or a random split
- Min_samples_splits: The number of samples needed for a node to split further
- Min_samples_leaves: The minimum number of samples that must be in each leaf node

The best settings for each parameter were:

- Max depth: None
- Criterion: friedman_mse
- Splitter: random
- Min samples split: 3
- Min samples leaf: 6
- Best $R^2$ score: 0.4766

The 5th model used ridge regression which is similar to multiple regression; however, it adds a penalty to the loss function to shrink coefficients and prevent overfitting. This was used due to possible multicollinearity in the dataset since we included the race share variables. Finally, model 6 was a random forest regressor.

**Results:**

Of the six models, the random forest regressor performed the best ($R^2 = 0.6624$ (Figure 4)). The next best was our decision tree regressor with hyperparameter tuning ($R^2 = 0.5248$). Then following that it was our ridge regression ($R^2 = 0.4633$), multiple regression model with race share variable ($R^2 = 0.4630$ (Figure 5)), multiple regression model without race share variables ($R^2 = 0.4183$ (Figure 6)), and basic decision tree regressor ($R^2 = 0.3099$). An $R^2$ value is essentially how far off our predicted points and actual points are from our line of best fit, where they typically range from 0 to 1 mimicking a percentage like score, when multiplied by 100. Based on this, the scores that we got do not seem very good, as you want something close to 100% and we got values around 50%, meaning our models explain about 50% of the variability of graduation rates. However, when trying to predict human behavior, like we are, we learned

that humans are very messy and unpredictable, and most of their data is all over the place. So we concluded that these scores were actually really good for our problem, as our best one predicted about 66% of the variability of graduation rates.

In order to determine if online schooling was beneficial to students we looked at the feature importance for 3 of the 6 models. We looked at the best scoring 2 (random forest regressor (Figure 7), and decision tree regressor with hyperparameter tuning (Figure 8)) and the worst model (basic decision tree regressor (Figure 9)) to see what was impacting our models. By not just using the best models and also looking at the worst allows us to be able to conclude that they are truly impactful. Our results were very similar across all three. The number one factor for all three was total_enrollment (40%), which was just a given as the more people at a given university, the higher the graduation rates will be. So then, we looked at the next most impactful variable which was also the same for all three, average net price (20%), which as we stated earlier represents average net cost after financial aid. This tells us that the more expensive a university is the higher the graduation rate they will have. This also makes sense as schools with more money will be able to have more resources and better programs compared to those who are not as fortunate when it comes to money. Our third most impactful variable was black_share (15%), or proportion of enrolled students who are black. This was very surprising to see race so high on our importance charts, as we would think these would be much lower as race should not impact a school's graduation rate. However, when looking back at the dataset we discovered that since we used only public universities in the US, over a third of them (35%) were HBCUs, which caused the skewed results to show black_share as such a high importance factor. Lastly, the fourth most important factor for 2 of the 3 was distance_only_share_imputed, which again is the imputed proportion of students enrolled exclusively in distance education coming in at around 8%. Seeing this so low on the important chart, led us to conclude that the percentage of distance learners had a measurable, but not significant impact on graduation rates.

**Conclusion and Limitations:**

Through synthesis of various models and exploratory data analysis, we have concluded that an increase of distance learning as a medium does not significantly influence a given student’s likelihood of graduating, given the variables we chose to explore. Despite these

findings, we are aware of improvements that could have been made to the dataset to account for other variables, such as student test scores, post-graduate job performance, or other metrics that would capture a more granular picture. We also are aware of how only using students who are fully remote learners, and not using students who take a mix of in-person and remote classes could have also improved our dataset and findings.

**AI Acknowledgement & Resources**

Portions of code were copied (and then edited) from previous in-class studio notes/assignments as well as the websites listed below. Additionally, GitHub Copilot was occasionally used to write code with its Inline Suggestions feature.

ChatGPT was used at the end of our write up to fix grammatical errors and spelling.

Data source:

- Multiple Regression: https://www.w3schools.com/python/python_ml_multiple_regression.asp & Final Project from DTSC 1302
- Decision Tree Regressor: https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html
- Ridge Regression: https://www.geeksforgeeks.org/machine-learning/what-is-ridge-regression/ & https://medium.com/@dvdhartsman/extracting-coefficients-and-feature-names-from-scikit-learn-pipelines-331d578b8450
- Random Forest Regressor: https://www.geeksforgeeks.org/machine-learning/random-forest-regression-in-python/

**Figures:**

Figure 1:

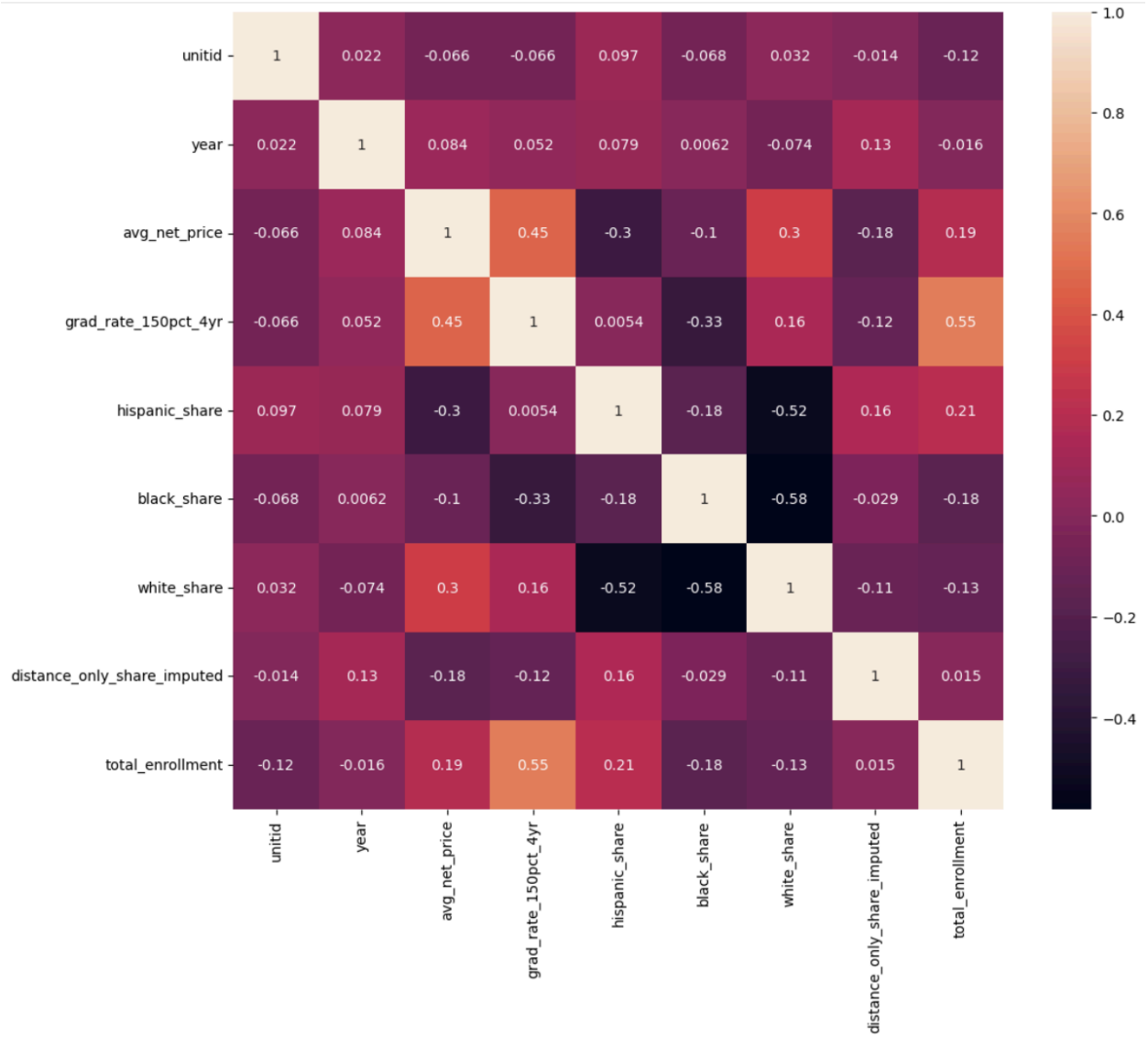



Figure 2:

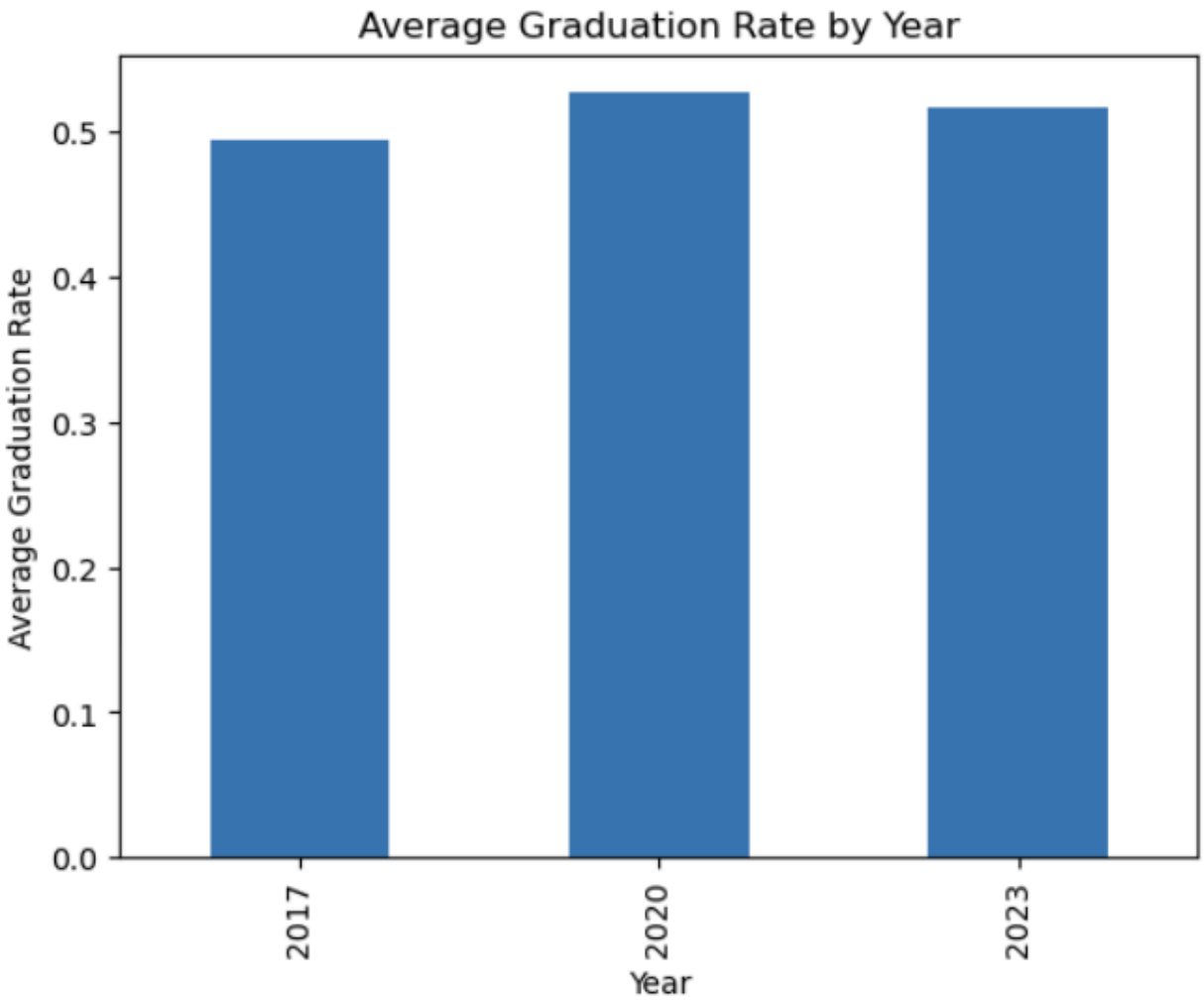

Average Graduation Rate by Year
Average Graduation Rate
0.0
0.1
0.2
0.3
0.4
0.5
2017
2020
2023
Year

Figure 3:

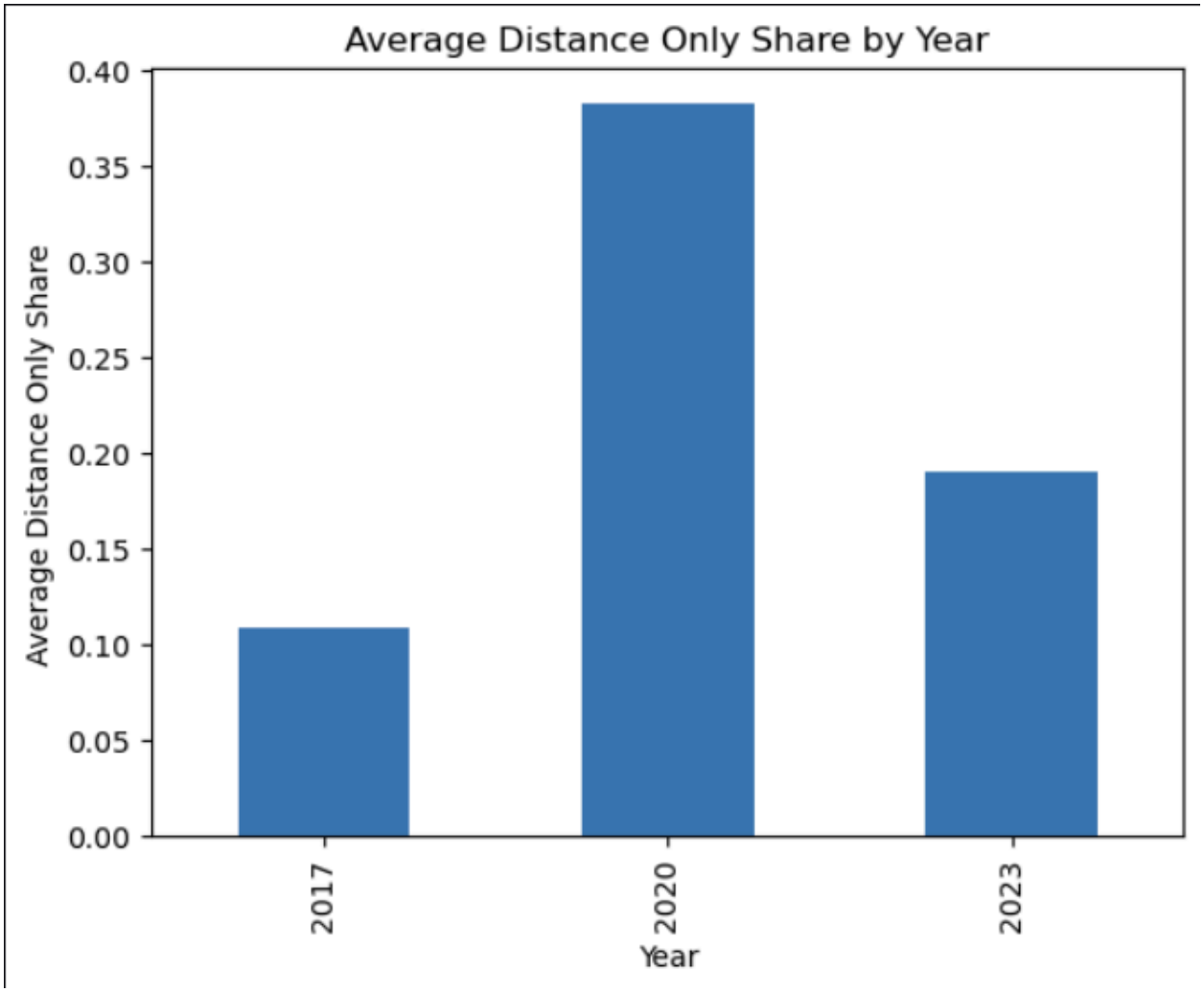

Average Distance Only Share by Year
Average Distance Only Share
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
2017
2020
2023
Year

Figure 4:

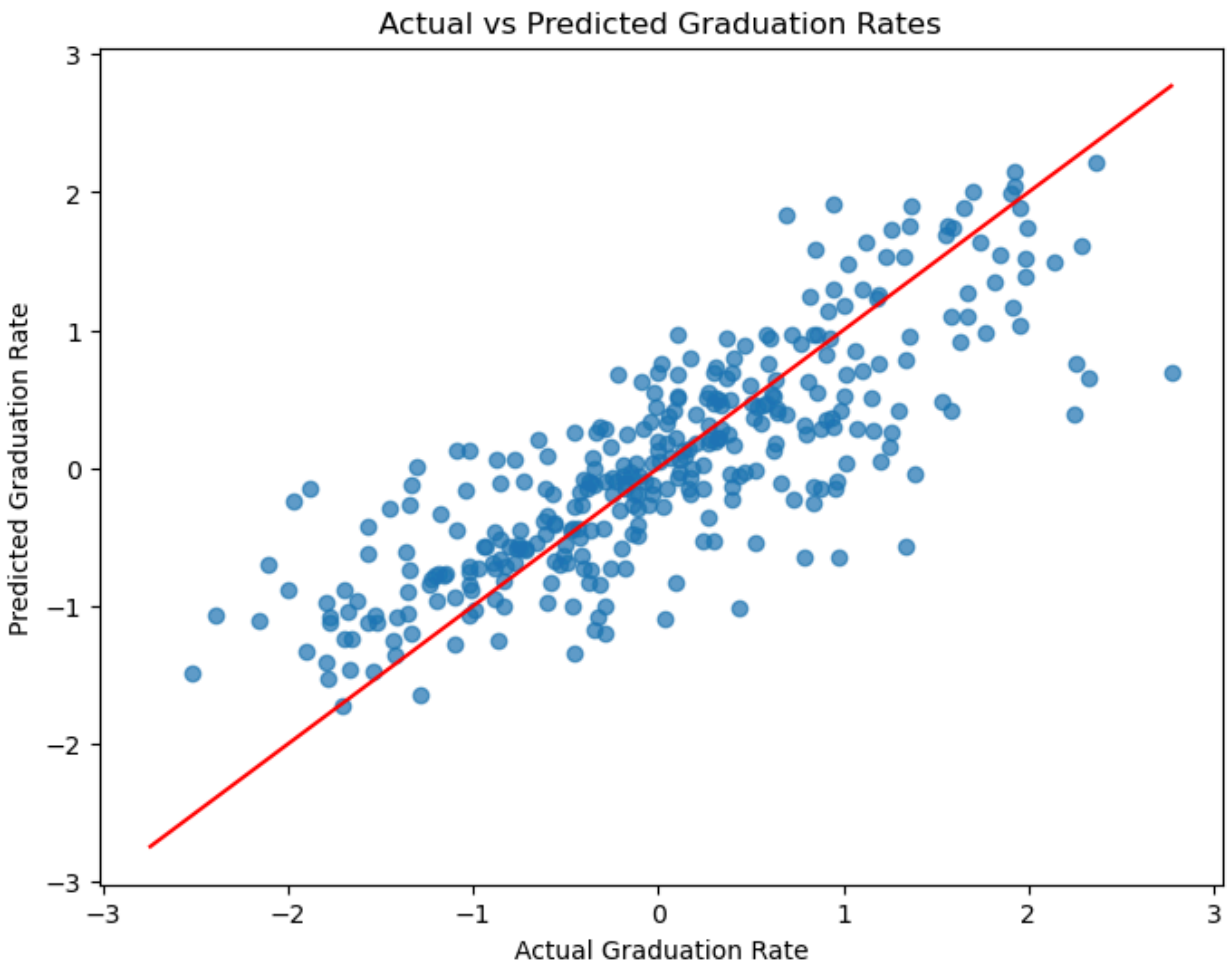



Figure 5:

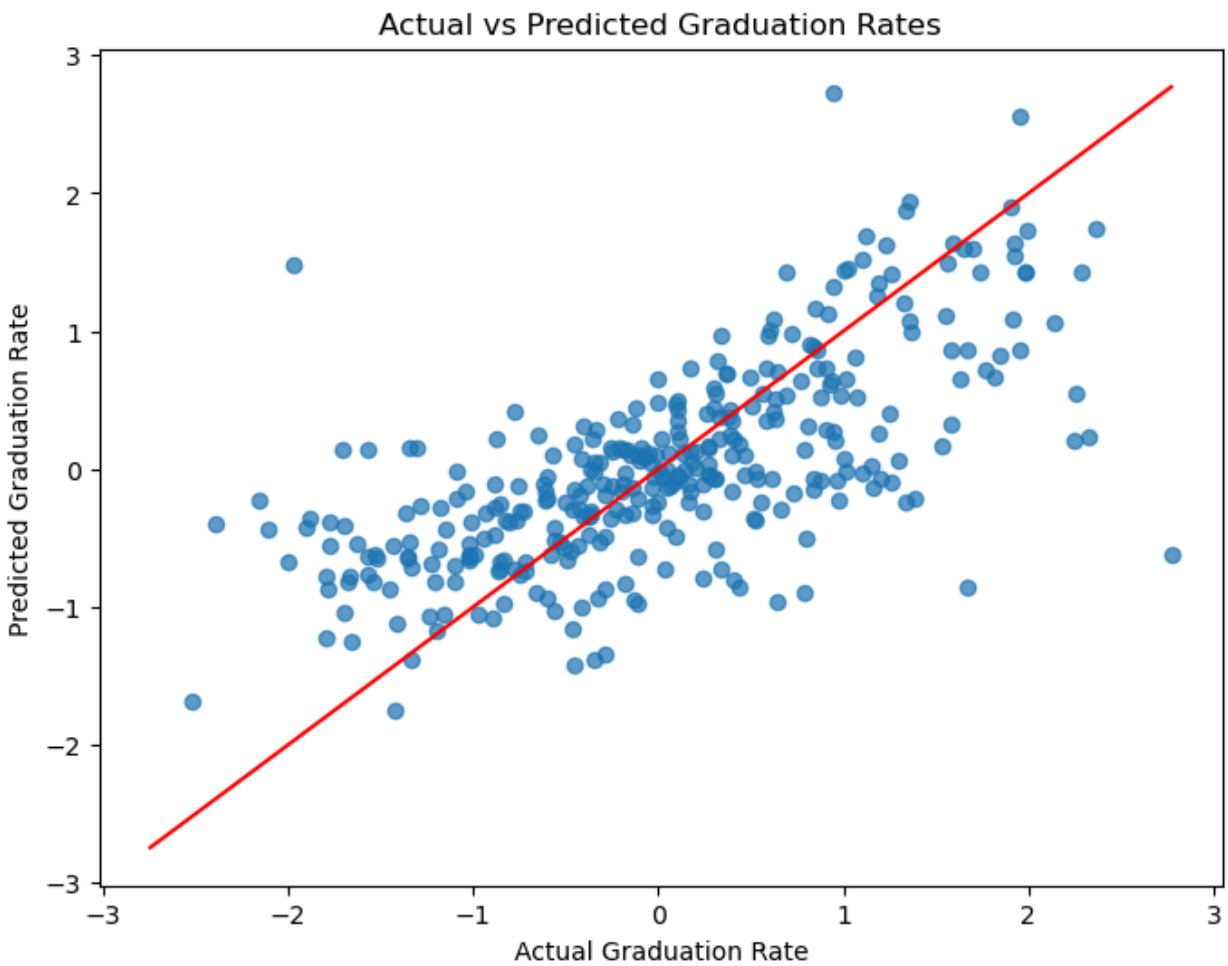

Actual vs Predicted Graduation Rates
Predicted Graduation Rate
Actual Graduation Rate
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

Figure 6:

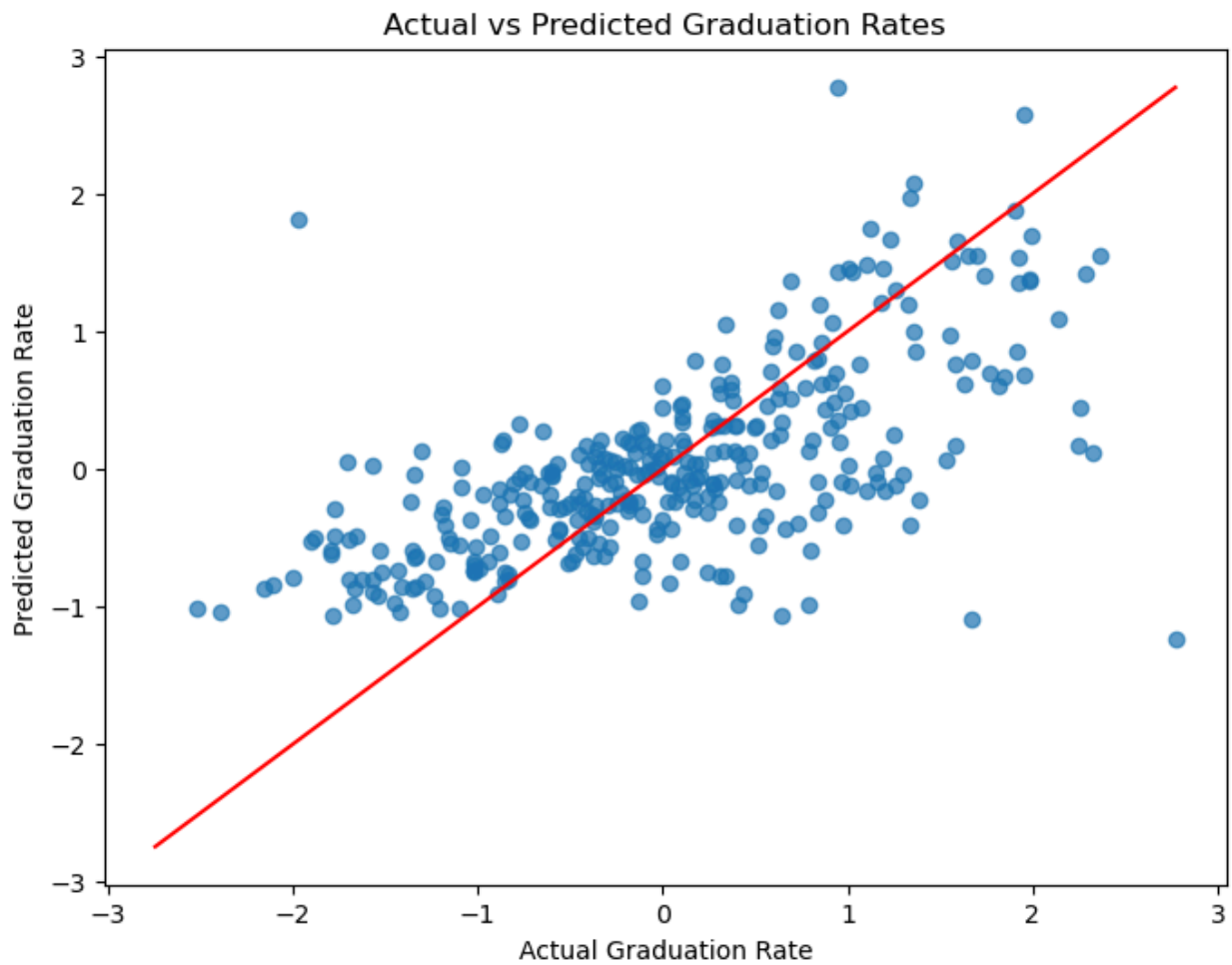



Figure 7:

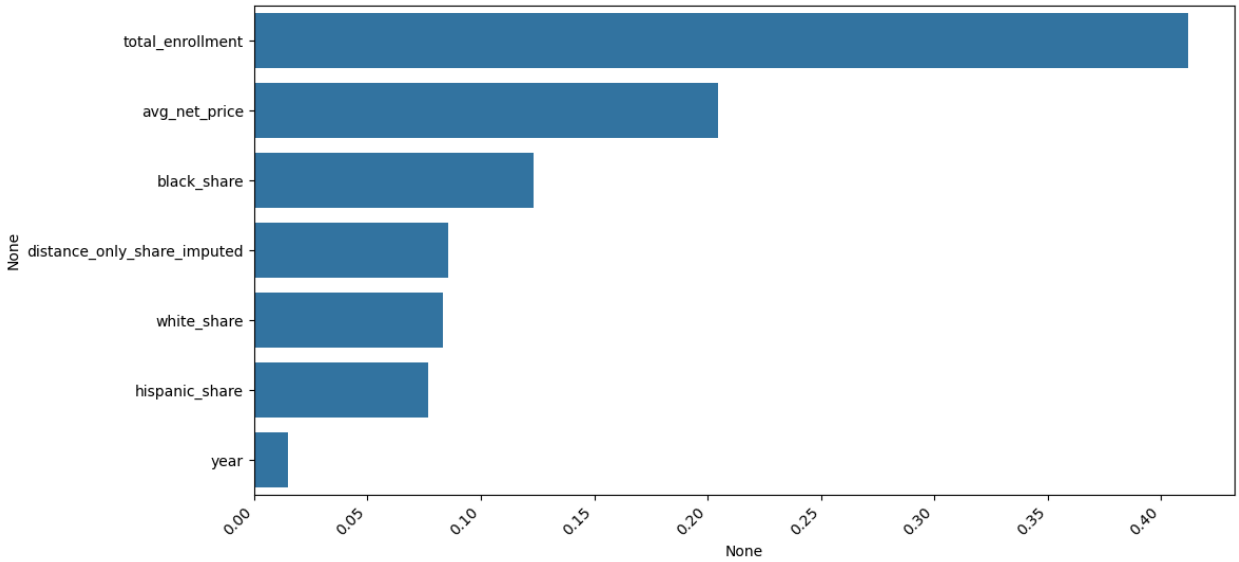
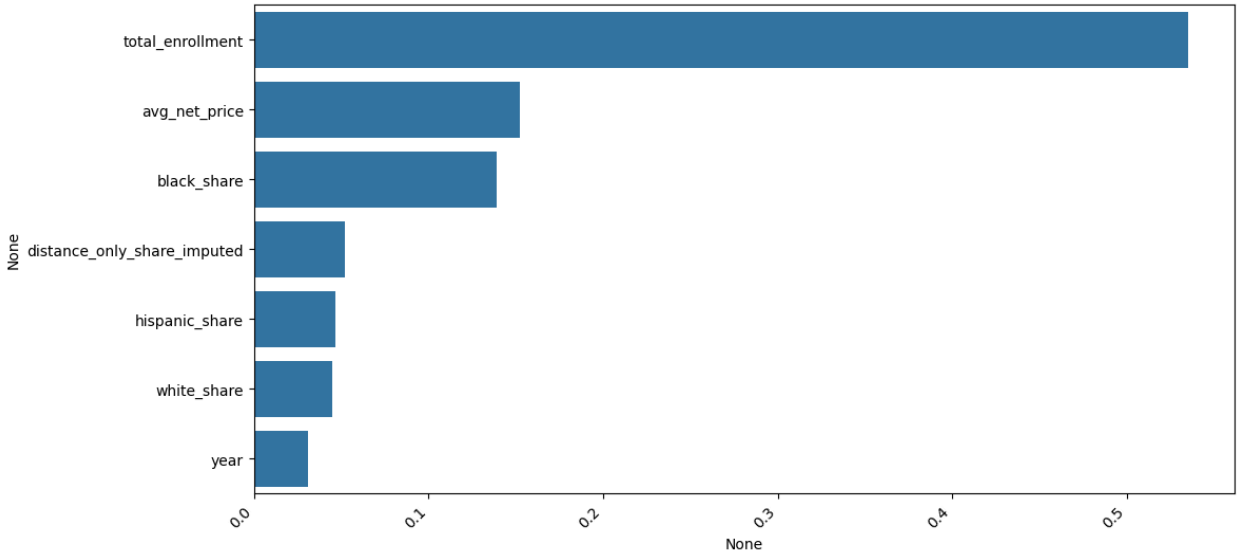



Figure 8:


total_enrollment
avg_net_price
black_share
distance_only_share_imputed
hispanic_share
white_share
year
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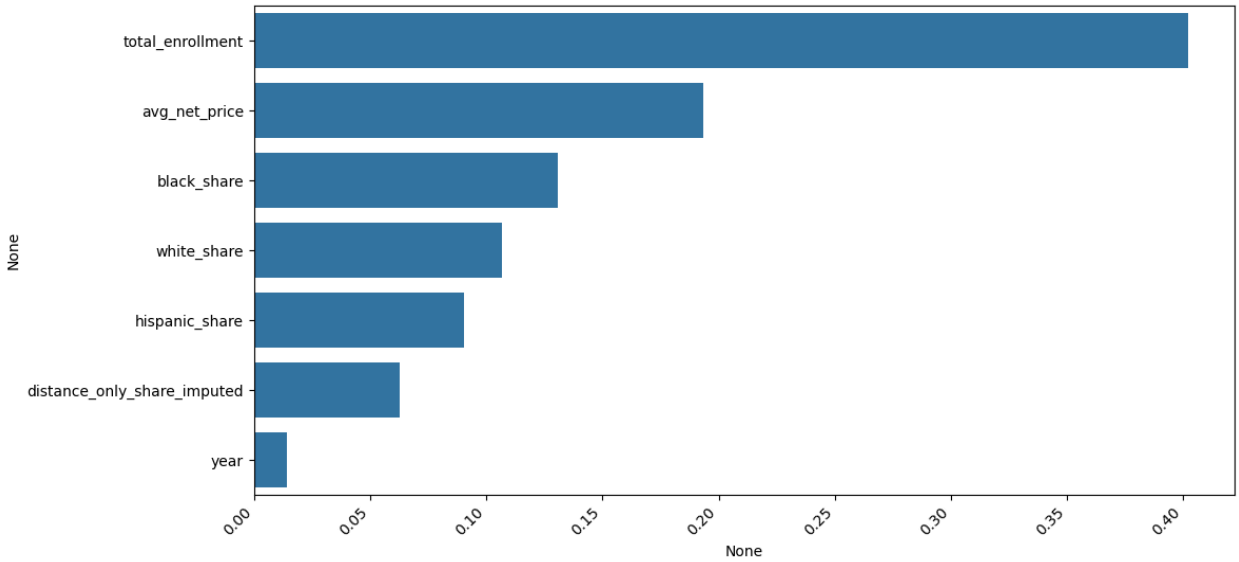
0.0
0.1
0.2
0.3
0.4
0.5
None

Figure 9:


total_enrollment
avg_net_price
black_share
white_share
hispanic_share
distance_only_share_imputed
year
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0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
None